INDEX

M1 – Method mark, awarded for correct mathematical technique, independent of numbers used.

A1 – Answer mark, awarded for a correct result to a calculation, dependent on M

B1 – Independent mark, can be awarded regardless of other marks, for a correct method

E1 – Explanation mark, for correct use of terminology and mathematics for reasoning

F – Follow through from an incorrect result

AWRT – Answers which round to a given number

AWFW – Answers which fall within a given range

ACF – Any correct form, answers which are mathematically equivalent to the given answer

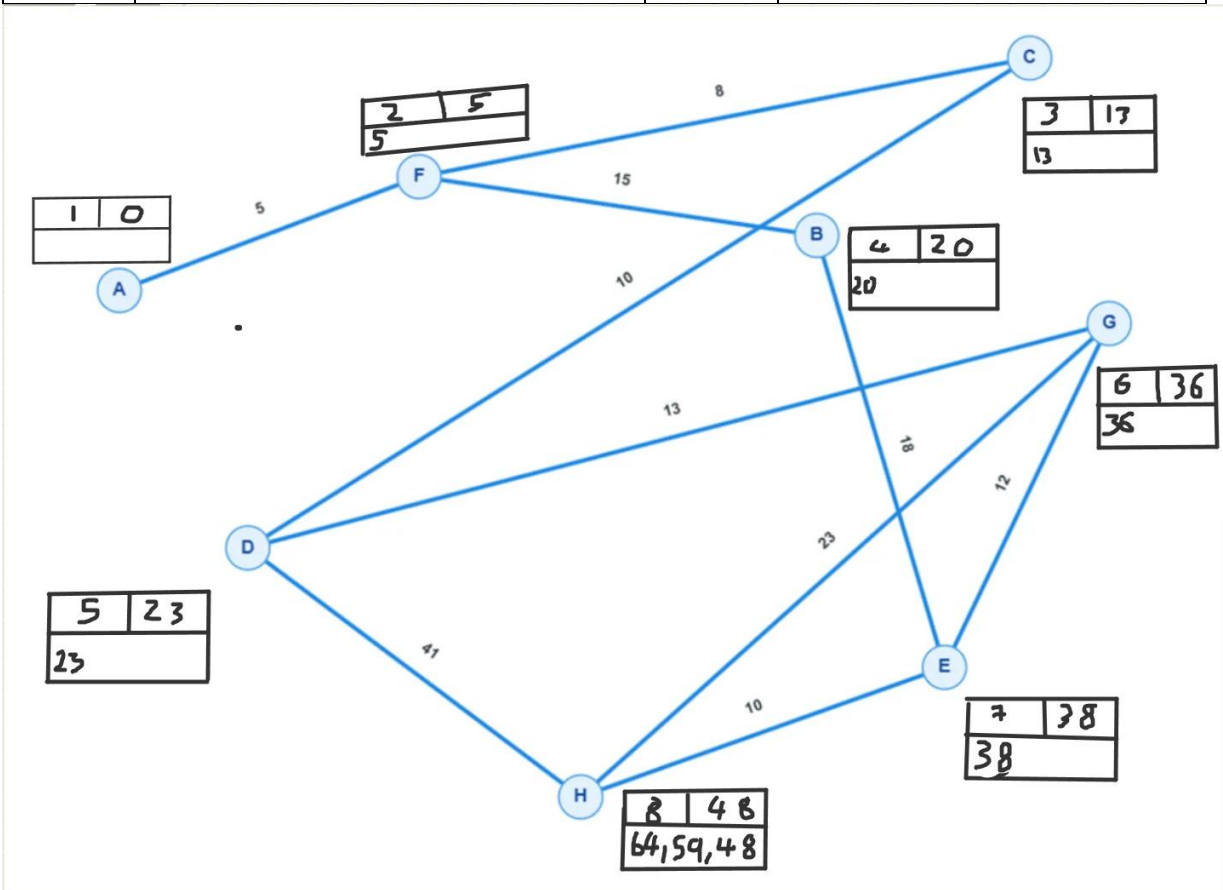
OE – Or Equivalent, numbers expressed in different forms

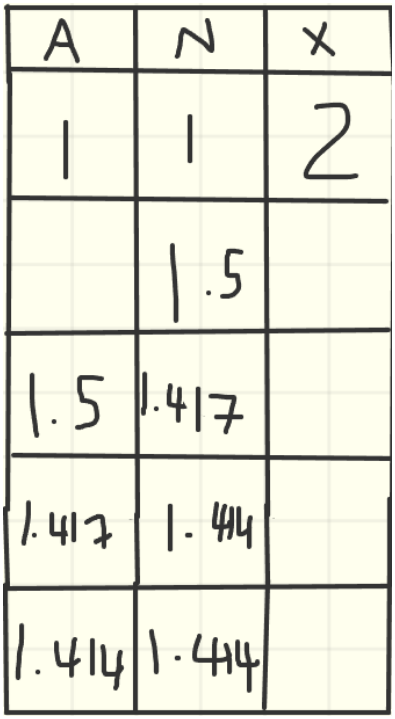
NE – not enough, not awardable/insufficient

A candidate may only lose a maximum of one mark throughout the paper for omission of units

Q	Marking Instructions	Marks	Notes
1 (i)	1 and 3	**A1**	**Do not award if any more nodes are mentioned**
	Sub Total	**1**	
1 (ii)	(A) $(A)\begin{pmatrix} 2 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 2 & 1 & 1 & 0 \\ 1 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 2 \end{pmatrix}$	**A1**	**Accept variations, given rows and columns are suitably labelled. This matrix has the top left index as 1,1 with the bottom right as 7,7**
	(B) M represents the number of ways to reach one node from another **via one arc**	**E1**	**NE "ways to get from one node to another",**
	Sub Total	**2**	
1 (iii)	(A) This is a connected graph, since there is at least one arc connecting each node to another.	**E1**	**Accept reference to all nodes having a path to another node, or reference to the incidence matrix**
	(B) One arc from 3 to 4 removed and the loops at 1 and 7 removed	**A1**	**If any more removed, award 0**
	Sub Total	**2**	
	Question Total	**5**	

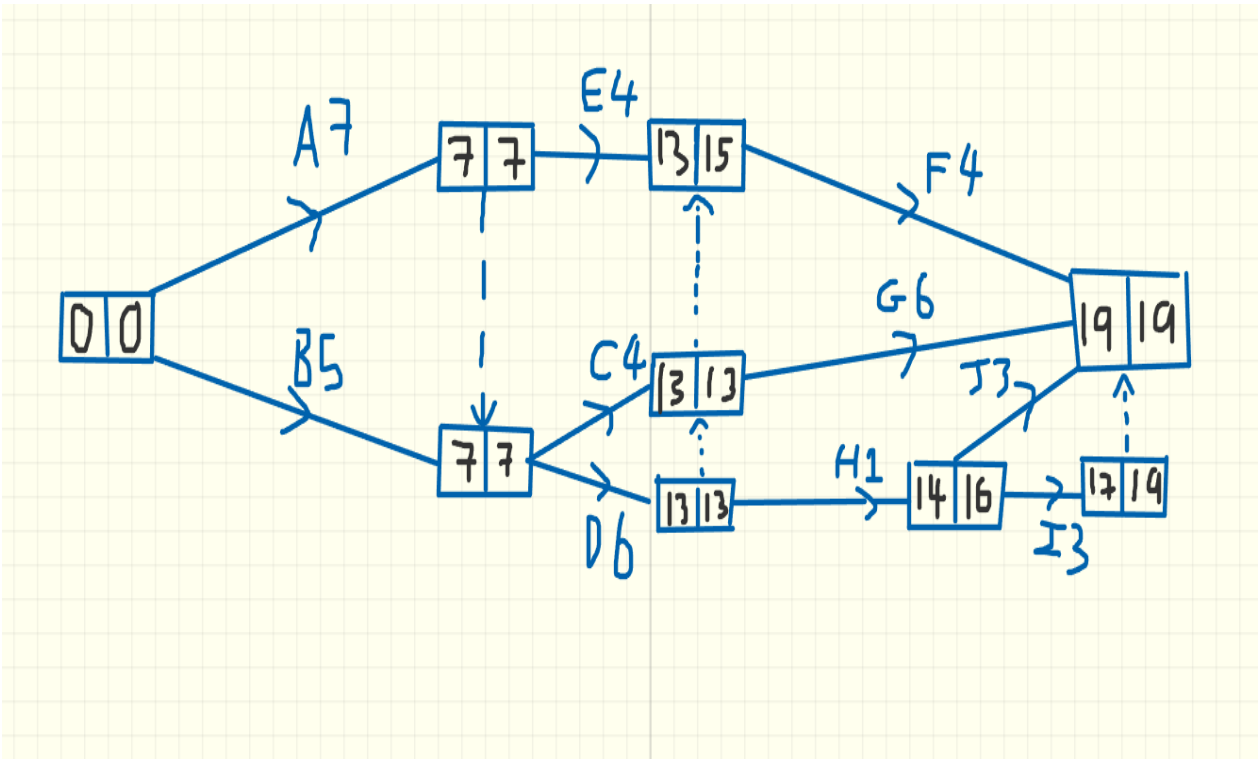
Q	Marking Instructions	Marks	Notes
2 (i)	(A) Diagram below (B) Minimum path - CFAFBE Weight - 51	**M1** **M1** **M1** **A1** **A1** **A1** **A1**	**For 2 correctly labelled** **For 5 correctly labelled** **For all correctly labelled** **Minimum path** **AFBEH** **Weight of 48**
	Sub Total	**7**	
2 (ii)	48	**A1**	
	Sub Total	**1**	
	Question Total	**8**	



Q	Marking Instructions	Marks	Notes
3 (i)	 Output 1.414	**A1** **A1** **A1**	**All values of X recorded correctly** **All values of N recorded correctly** **All values of A recorded correctly.**
	Sub Total	**3**	
3 (ii)	Letting X be negative causes N to become negative, since X/A is always smaller than A. This causes $\lvert A - N \rvert$ to always be greater than 0.001, meaning the algorithm never terminates	**E1** **E1 dep**	**Accept reference to the purpose of the algorithm, being to estimate the square root of X, with valid justification of a square root of a negative being non-real**
	Sub Total	**2**	
	Question Total	**5**	

Q	Marking Instructions	Marks	Notes
4 (i)	**13** 30 22 26 29 32 59 12 11 38 27 33 **12** 11 13 **30** 22 26 29 32 59 38 27 33 **11** 12 13 **22** 26 29 27 30 **32** 59 38 33 11 12 13 22 **26** 29 27 30 32 **59** 38 33 11 12 13 22 26 **29** 27 30 32 **38** 33 59 11 12 13 22 26 **27** 29 30 32 **33** 38 59 11 12 13 22 26 27 29 30 32 33 38 59	**M1** **M1** **A1 dep**	**First pass correct.** **The correct use of the first element in each sublist as a pivot** **Correctly sorted list in ascending order, with all passes correct**
	Sub Total	**3**	
4 (ii)	Tube 1: 59 Tube 2: 38 22 Tube 3: 33 27 Tube 4: 32 26 Tube 5: 30 29 Tube 6: 13 12 11	**M1** **A1**	**Any bin correct** **All bins correct**
	Sub Total	**2**	
4 (iii)	$332 = \sum tube\ lengths$ $\frac{332}{60} \approx 5.533$, therefore 6 bins are needed My answer to part (ii) is optimal.	**E1**	**The calculation must be seen,**
	Sub Total	**1**	
	Question Total	**6**	

Q	Marking Instructions	Marks	Notes
5 (i)	Diagram below	**M1** **A1** **A1**	**A and B from a single start vertex, with E directly from A, and C or D directly from B** **F from E, G from C, H from D, with J and I from H. Award if I and J are swapped.** **Four dummy activities correctly placed and directed suitably. Do not award if more are used.**
	Sub Total	**3**	
5 (ii)	(A) Diagram below (B) 19 hours (C) ADG	**M1** **A1** **A1** **A1** **A1**	**At least 3 event times labelled** **All early event times correct** **All Late event times correct** **Do not award if hours missing** **Accept any order of stating**
	Sub Total	**5**	
5 (iii)	(A) $16 - 13 - 1 = 2\ hours$ (B) If H is delayed by 5 hours, the 2-hour float is used, then 3 more hours is wasted, therefore H starts at 18, finishing at 19. The project will then finish after J and I, so finish after 22 hours, which is 3 hours late. $1200 - 3 \times 50 = \$1050$	**M1** **A1** **M1 (or R1)** **A1**	**Attempt to calculate independent or total float** **Award if hours missing** **Attempt to find new finish time for project** **Accept without units**
	Sub Total	**4**	
	Question Total	**12**	


A7
E4
F4
B5
C4
G6
D6
H1
J3
I3
0 0
7 7
7 7
13 15
13 13
13 13
14 16
17 19
19 19

Q	Marking Instructions	Marks	Notes
6 (i)	From the time constraint, 1.5 hrs = 90 mins $9x + 6y + 18z \leq 90$ $3x + 2y + 6z \leq 30$ Discrete at least twice Core Pure, $x \geq 2z$ Statistics is at least Discrete, $y \geq x$ Maximise $P = 4x + 3y + 9z$ Subject to: $3x + 2y + 6z \leq 30$ $2z - x \leq 0$ $x - y \leq 0$ $x \geq 0, y \geq 0, z \geq 0$ $x, y, z \; are \; integers$	**M1** **M1** **M1** **A1** **A1**	**One constraint, justified** **A different constraint (justified)** **Either constraint simplified properly** **Objective function with Maximise** **All constraints correct, and simplified, allow x-2z, y-x, given the inequality sign is correct**
	Sub Total	**5**	
6 (ii)	(A) Let s_1, s_2, s_3 be slack variables $3x + 2y + 6z + s_1 = 30$ $2z - x + s_2 = 0$ $x - y + s_3 = 0$ $P - 4x - 3y - 9z = 0$ Tableau Below (B) z is most negative 45/6 = 7.5, 0/2 = 0, 0/0 = ignore Tableau Below	**M1** **M1** **A1** **A1** **M1** **M1** **A1**	**Slack variables defined** **One modified constraint** **Objective row correct** **Constraint rows correct** **Ratio tests with the z column. Assume from correct tableau** **One row correct.** **Correct tableau** **FT their initial tableau.**
	Sub Total	**7**	
6 (iii)	New constraint: $x \geq 5$ The standard Simplex algorithm is not able to handle greater than (or equal to) inequalities, as it starts the algorithm from the origin.	**E1**	**Constraint must be seen** **Accept that the new constraint means the origin is not a feasible solution**
	Sub Total	**1**	
	Question Total	**13**	

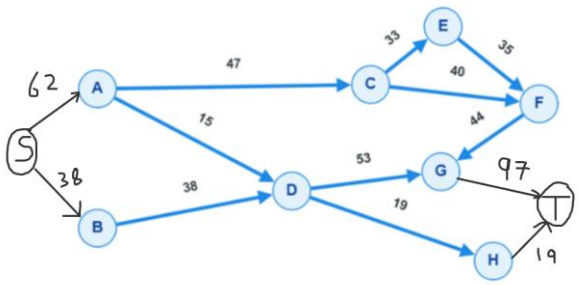
This LP has a boring solution, 15 stats questions 😟

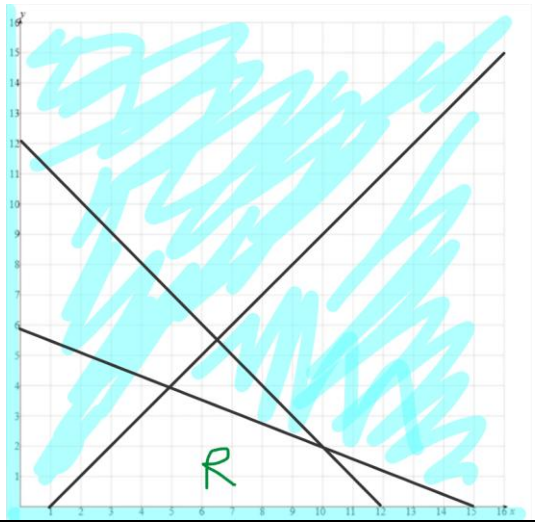
(A)

P	x	y	z	S_1	S_2	S_3	RHS
1	-4	-3	-9	0	0	0	0
0	3	2	6	1	0	0	30
0	-1	0	2	0	1	0	0
0	1	-1	0	0	0	1	0

(B)

P	x	y	z	S_1	S_2	S_3	RHS
1	$-\frac{17}{2}$	-3	0	0	$\frac{9}{2}$	0	0
0	6	2	0	1	-3	0	30
0	$-\frac{1}{2}$	0	1	0	$\frac{1}{2}$	0	0
0	1	-1	0	0	0	1	0

Q	Marking Instructions	Marks	Notes
7 (i)	(A)  (B) The sum of the source arcs is 100, the sum of the arcs going into T is 116, therefore both GT and HT cannot be saturated, since the flow into the network from the source must equal the flow into the sink	**A1** **E1**	**Valid justification, do not accept any reference to cuts of the network**
	Sub Total	**2**	
7 (ii)	(A) Capacity $\alpha = 97$ (B) Capacity $\beta = 191$ (C) Via the max-flow min-cut theorem, the max flow is 97. 97 is greater than 83, therefore this is a suitable network for the client	**A1** **A1** **A1**	**f.t their α & β, do not award if not justified**
	Sub Total	**3**	
	Question Total	**5**	

Q	Marking Instructions	Marks	Notes
8 (i)	$z = 24 - x - y$ $2x + 24 - x - y \geq 25$ $-y + x \geq 1$ $24 - x - y \geq 12$ $x + y \leq 12$	**M1** **A1**	**Correct substitution established** **The two relevant constraints reformulated.** **f.t their substitution**
	Sub Total	**2**	
8 (ii)		**M1** **A1** **A1**	**One constraint correctly sketched** **All constraints correctly sketched** **Region clearly indicated**
	Sub Total	**3**	
8 (iii)	P=71	**A1**	
	Sub Total	**1**	
	Question Total	**6**	